THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The formula of calcium hydroxide is: 1

(a) Ca_2OH **(b)** $CaOH$ **(c)** $Ca(OH)_2$ **(d)** CaO_2H

2. Which element has a valency of 3? 1

(a) chlorine **(b)** sodium **(c)** magnesium **(d)** aluminium

3. The area of a rectangle 50 cm by 30 cm in square metres is: 1

(a) 1.5 **(b)** 15 **(c)** 0.015 **(d)** 0.15

4. Which of the following will pass through a filter paper? 1

(a) sand **(b)** colloidal particles

(c) chalk powder **(d)** iron filings

5. Which compound has a double bond between its atoms? 1

(a) N_2 **(b)** O_2 **(c)** H_2 **(d)** Cl_2

6. The isotopes of an element differ in the number of: 1

(a) neutrons **(b)** protons **(c)** electrons **(d)** shells

7. The volume of a cuboid 2 m x 50 cm x 20 cm in cubic metres is: 1

(a) 2 **(b)** 200 **(c)** 20 **(d)** 0.2

8. To make 500 g of a 6 per cent solution, the mass of water needed is: 1

(a) 500 g **(b)** 30 g **(c)** 470 g **(d)** 494 g

9. Carbon monoxide has C : O = 3 : 4. The mass of oxygen that combines with 9 g of carbon is: 1

(a) 6.75 g **(b)** 16 g **(c)** 9 g **(d)** 12 g

10. The ion Mg^{2+} has the electronic configuration: 1

(a) 2, 8 **(b)** 2, 8, 8 **(c)** 2, 6 **(d)** 2, 8, 2

11. When a model fails in some situation, the scientist: 1

(a) ignores the failure

(b) defines its limits and develops a better model

(c) discards all science

(d) blames observers

12. To recover pure sugar from sugar solution, one uses: 1

(a) decantation **(b)** sublimation **(c)** filtration **(d)** crystallisation

13. The formula of iron(III) oxide is: 1

(a) FeO **(b)** Fe_2O_3 **(c)** Fe_3O_2 **(d)** Fe_2O

14. Which of the following has the maximum valency? 1

(a) fluorine (2, 7) **(b)** oxygen (2, 6) **(c)** carbon (2, 4) **(d)** sodium (2, 8, 1)

15. A scientist measures 12.5 g, 12.7 g and 12.6 g for the same mass. The mean is: 1

(a) 37.8 g **(b)** 12.5 g **(c)** 12.7 g **(d)** 12.6 g

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Calculate the percentage of carbon in CO_2. 2

17. The element X has valency 2 and its atom has 12 neutrons and mass number 24. Find the atomic number and the electronic configuration. 2

18. A student takes five readings of a mass: 5.1, 5.0, 5.2, 5.1, 5.1 g. Find the mean. 2

19. A solution is made by dissolving 12 g of salt in 88 g of water. Find the percentage by mass. 2

20. Calculate the mass of hydrogen and oxygen in 36 g of water. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Magnesium (4.8 g) combines with nitrogen to form 6.6 g magnesium nitride. Find the mass of nitrogen used and the ratio Mg : N. 3

22. Complete: ${}^{35}_{17}Cl$, ${}^{37}_{17}Cl$: number of p, n, e in each. What are they called? 3

23. Explain how an astronomer can find the distance to a star in light years and what a light year is. 3

24. Differentiate between a true solution and a colloid in three points with an example for each. 3

OR

Explain why an ionic solid does not conduct but its melt does. 3

25. Explain how radioactive iodine and cobalt help in medicine. 3

OR

State three safety rules for chemical laboratories and the reason for each. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the use of estimation in science with three worked examples and discuss the sources of uncertainty. 5

OR

Describe the separation of a mixture containing iron filings, ammonium chloride, salt and sand in correct order and explain the principle at each stage. 5

27. Explain how the electronic structure explains the formation of ionic compounds NaCl, MgO and CaF2 with the charges and the formulae. 5

OR

Chlorine has isotopes of 35 and 37 u. If its average atomic mass is 35.5 u find the percentage of each isotope, and explain the reason. 5

END OF QUESTION PAPER **Total: 50 Marks**

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